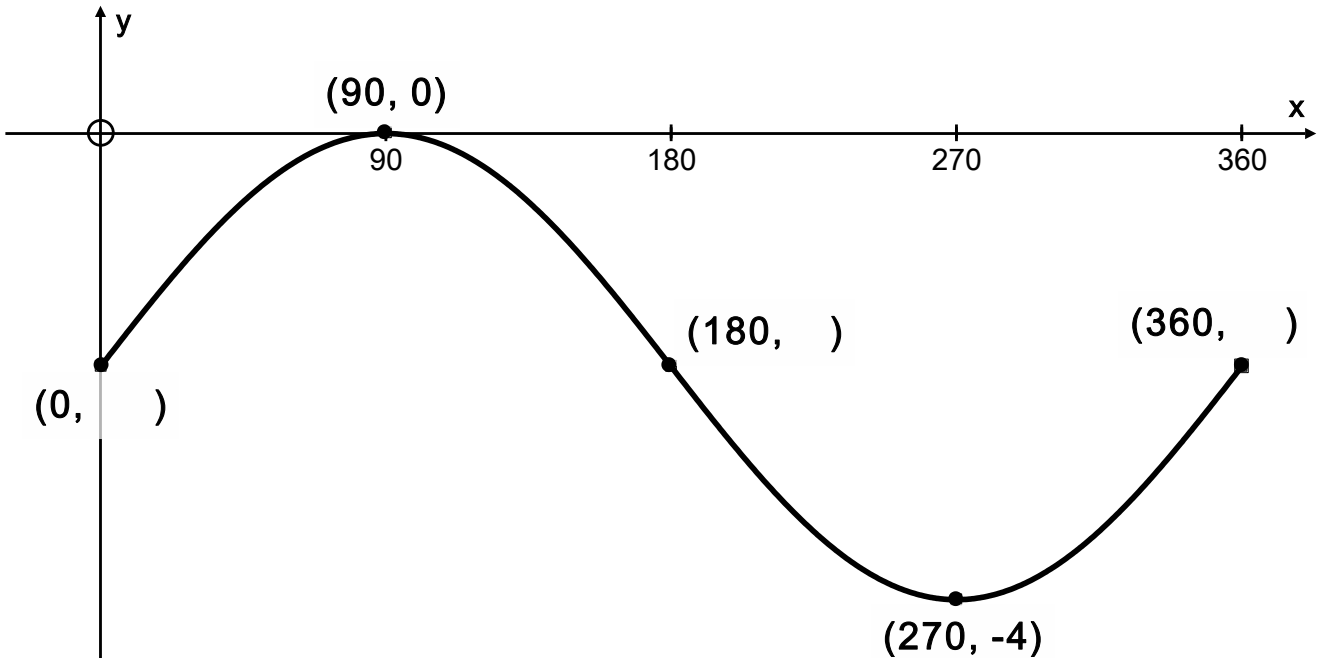
TRIG FUNCTIONS CONSOLIDATION MEMO

FINDING THE EQUATION OF TRIG GRAPHS

$$y = a\sin x + q$$

Example:

Sketched $f(x) = a \sin x + q$



Determine the values of *a* and *q*.

Working:

To find a: $|a| = \dfrac{Max\ value - Min\ value}{2} = \dfrac{0-(-4)}{2} = 2$

The sign of a : The graph is going up (increasing) from the y-intercept, $\therefore a > 0$

$\therefore a = 2$

To find q: $q = \dfrac{Max\ value + Min\ value}{2} = \dfrac{0+(-4)}{2} = \dfrac{-4}{2} = -2$

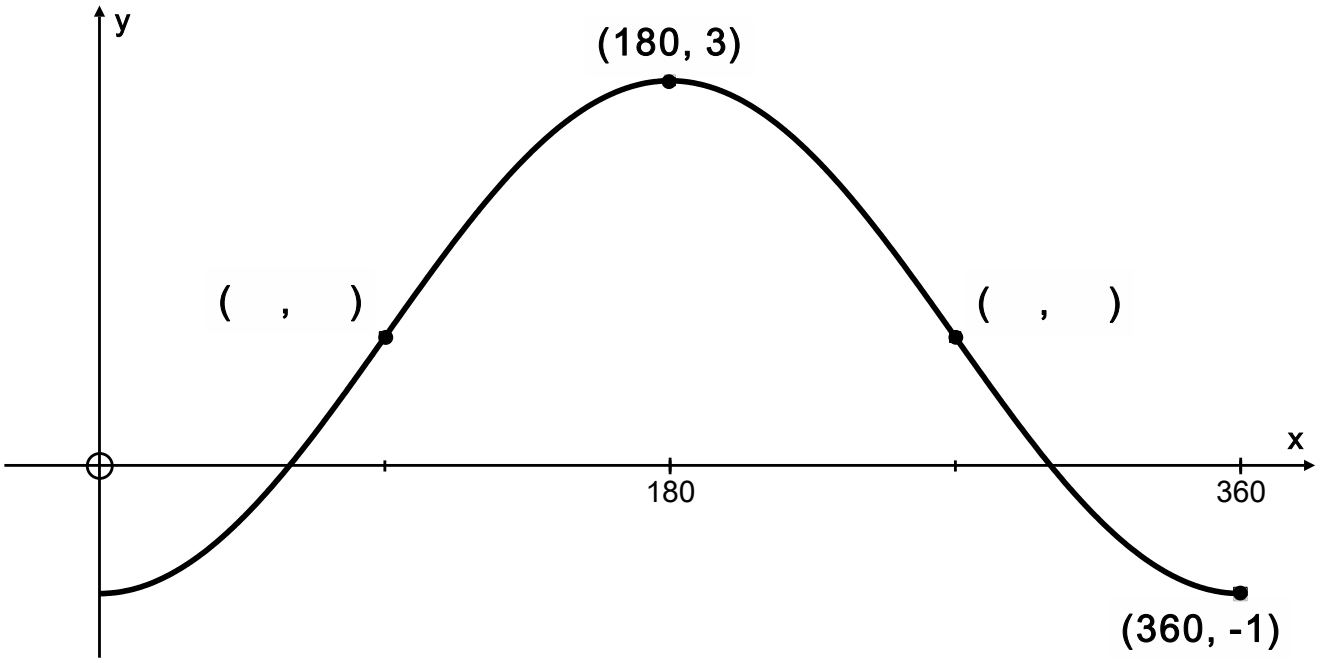
$\therefore q = -2$

$f(x) = 2\sin x - 2$

$$y = a\cos x + q$$

Example:

Sketched $f(x) = a\cos x + q$



Determine the values of *a* and *q*.

Working:

To find a: $|a| = \dfrac{Max\ value - Min\ value}{2} = \dfrac{3-(-1)}{2} = 2$

The sign of a : The graph is going up (increasing) from the y-intercept, $\therefore a < 0$

$\therefore a = -2$

To find q: $q = \dfrac{Max\ value + Min\ value}{2} = \dfrac{3+(-1)}{2} = 1$

$\therefore q = 1$

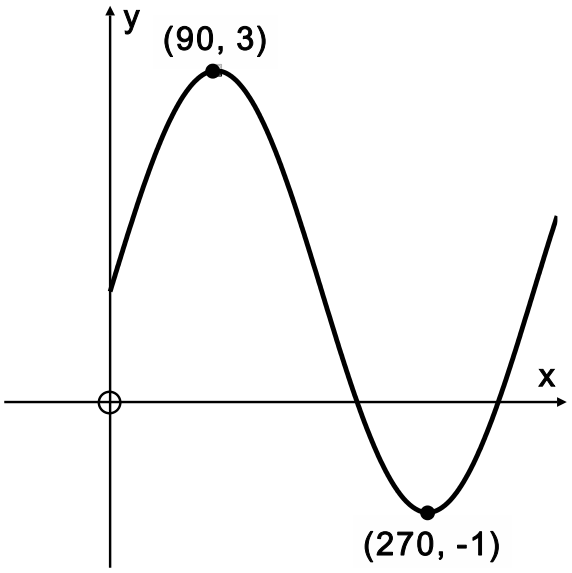
$f(x) = -2\cos x + 1$

FINDING THE EQUATION OF TRIG GRAPHS

Exercise A:

Find the values of *a* and *q* in each of the following:

1. $y = a\sin x + q$

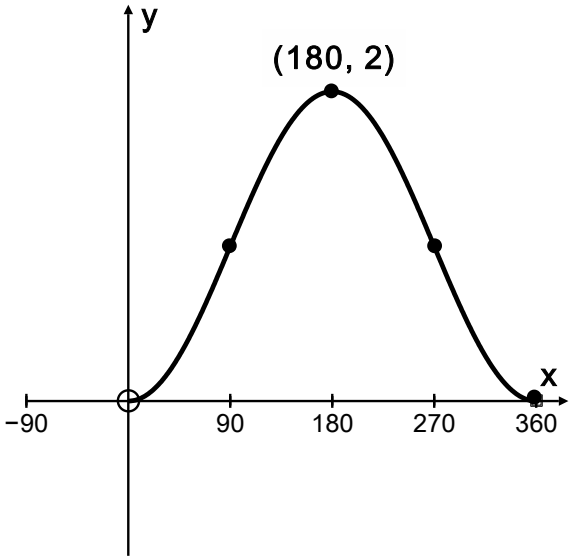


$a = 2$

$3 = 2\sin 90 + q$

$q = 1$

2. $y = a\cos x + q$

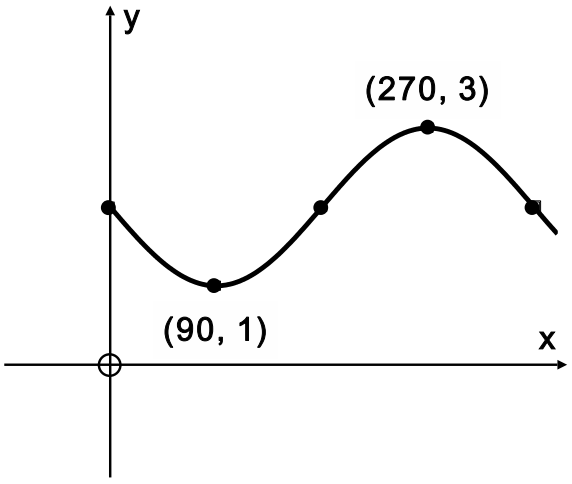


$a = -1$

$2 = \cos 180 + q$

$q = 1$

3. $y = a\sin x + q$

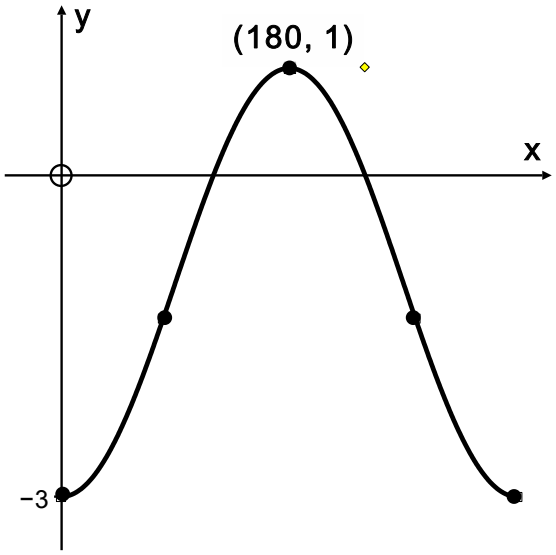


$a = -1$

$1 = -\sin 90 + q$

$q = 2$

4. $y = a\cos x + q$

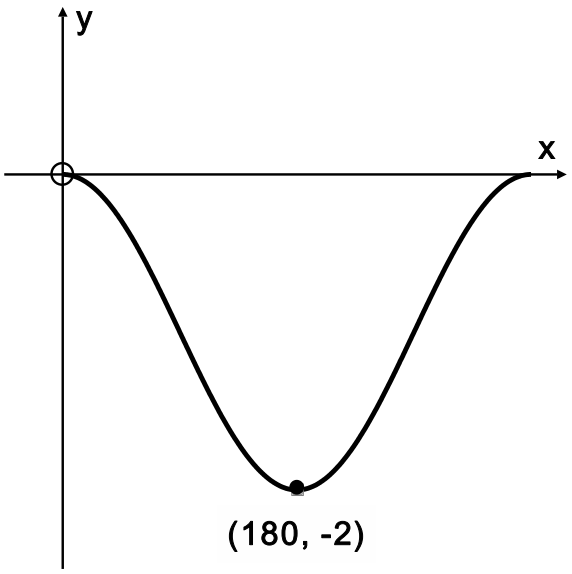


$a = -2$

$1 = -2\cos 180 + q$

$q = -1$

5. $y = a\cos x + q$

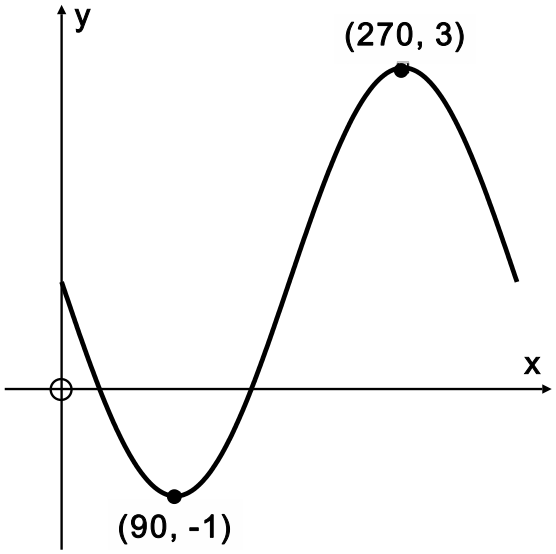


$a = 1$

$-2 = \cos 180 + q$

$q = -1$

6. $y = a\sin x + q$

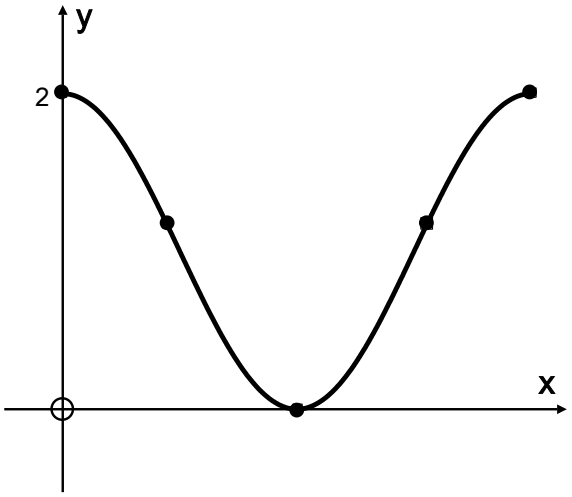


$a = -2$

$-1 = -2\sin 90 + q$

$q = 1$

7. $y = a\cos x + q$

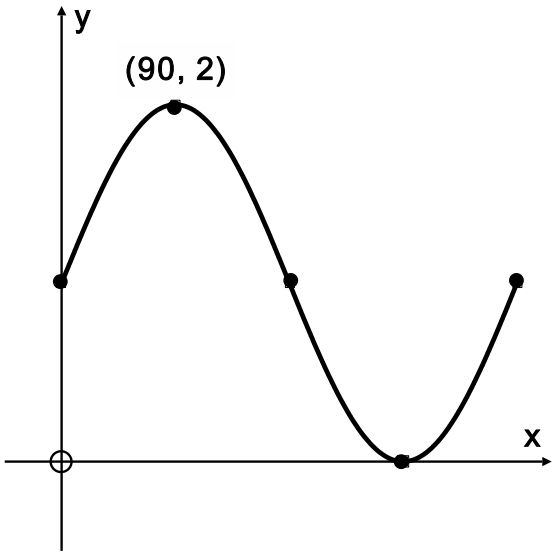


$a = 1$

$2 = \cos 0 + q$

$q = 1$

8. $y = a\sin x + q$



$a = 1$

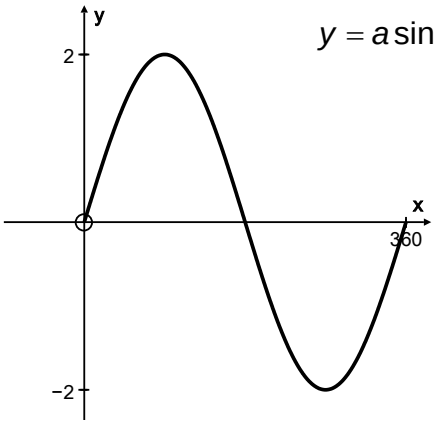
$2 = \sin 90 + q$

$q = 1$

Exercise B:

Find the values of a and q in each of the following:

1.



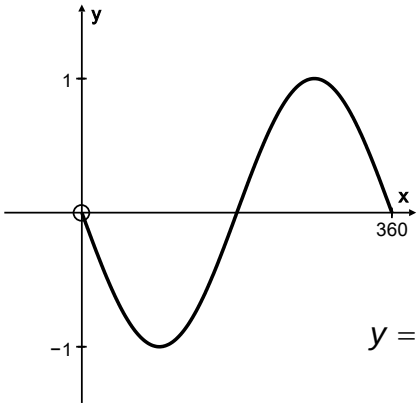
$y = a\sin x + q$

$a = 2$

$0 = 2\sin 0 + q$

$q = 0$

2.



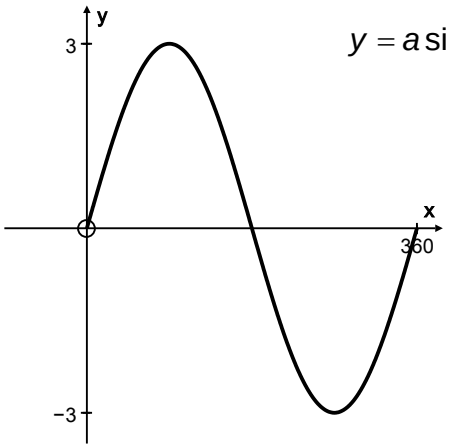
$y = a\sin x + q$

$a = -1$

$0 = -1\sin 0 + q$

$q = 0$

3.



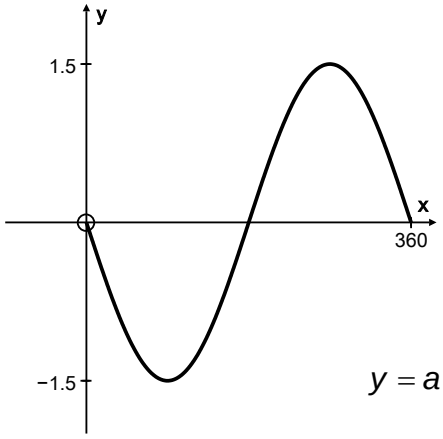
$y = a\sin x + q$

$a = 3$

$0 = 3\sin 0 + q$

$q = 0$

4.



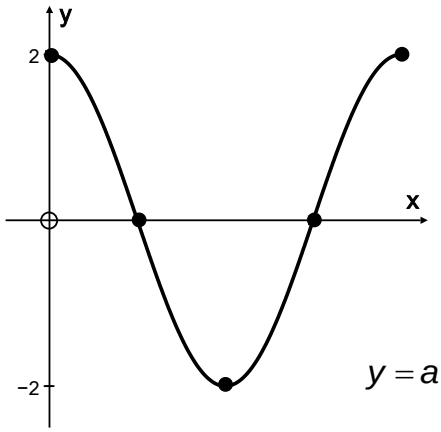
$y = a\sin x + q$

$a = -1{,}5$

$0 = -1{,}5\sin 0 + q$

$q = 0$

5.



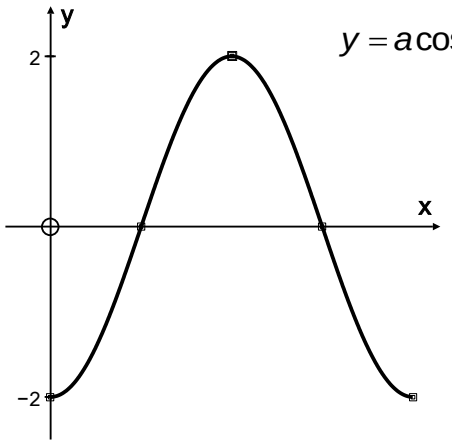
$y = a\cos x + q$

$a = 2$

$2 = 2\cos 0 + q$

$q = 0$

6.



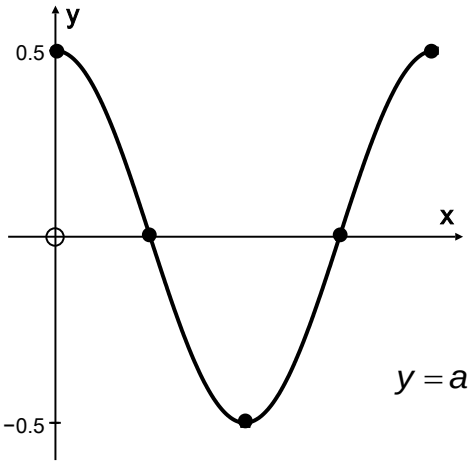
$y = a\cos x + q$

$a = -2$

$-2 = -2\cos 0 + q$

$q = 0$

7.



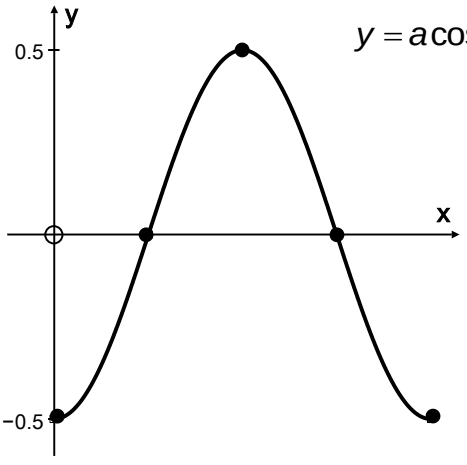
$y = a\cos x + q$

$a = \frac{1}{2}$

$\frac{1}{2} = \frac{1}{2}\cos 0 + q$

$q = 0$

8.



$y = a\cos x + q$

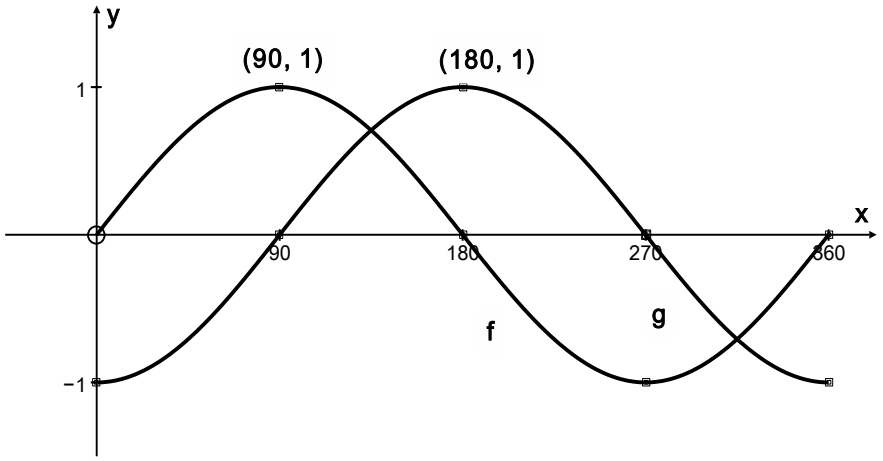
$a = -\frac{1}{2}$

$-\frac{1}{2} = -\frac{1}{2}\cos 0 + q$

$q = 0$

TRIG GRAPHS WORKSHEET (1)

1. Given: $f(x) = a\sin x$ *and* $g(x) = b\cos x \quad x \in [0^\circ;360^\circ]$



Period of f: 360°

Period of g: 360°

Amplitude of f: 1

Amplitude of g: 1

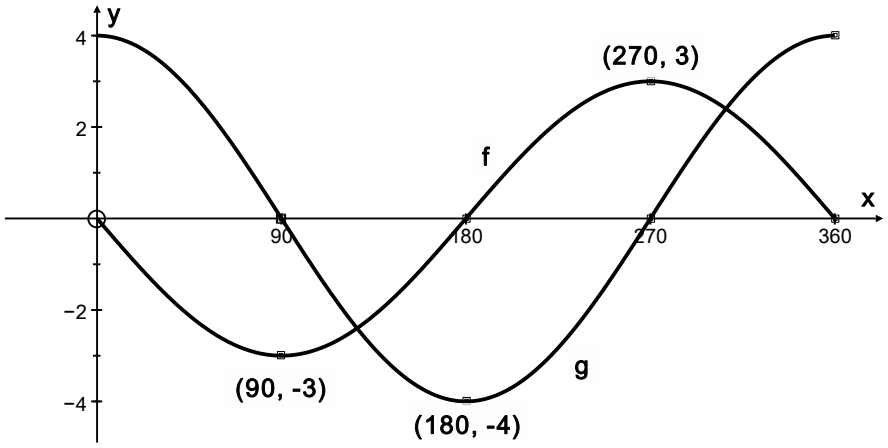
a) Find the values of a and b

$a = 1 \qquad b = 1$

b) For which values of x *is* $f(x) = g(x)$ *if* $x \in [0^\circ;360^\circ]$

$x = 135 \qquad x = 315$

2. Given: $f(x) = a\sin x$ *and* $g(x) = b\cos x \quad x \in [0^\circ;360^\circ]$



Period of f: 360°

Period of g: 360°

Amplitude of f: 3

Amplitude of g: 4

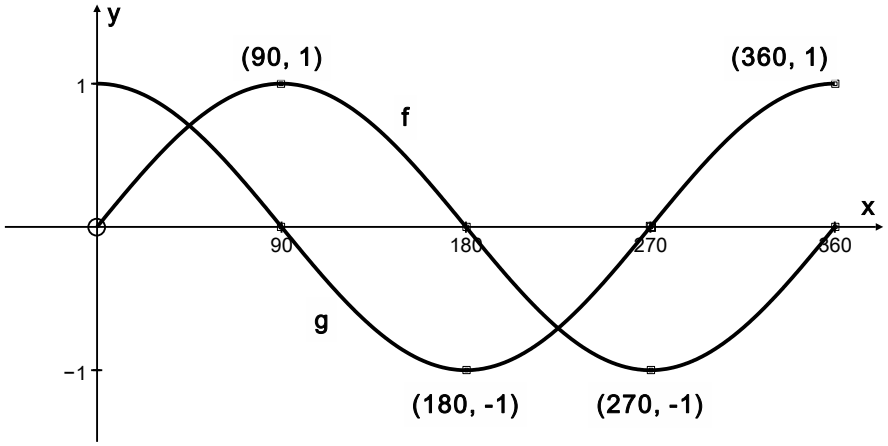
a) Find the values of a and b

$a = -3 \qquad b = 4$

b) For which values of x *is* $f(x) = g(x)$ *if* $x \in [0^\circ;360^\circ]$

$x = 126{,}9^\circ \qquad x = 306{,}9^\circ$

3. Given: $f(x) = a \sin x$ *and* $g(x) = b\cos x \quad x \in [0^\circ; 360^\circ]$



Period of f: 360°

Period of g: 360°

Amplitude of f: 1

Amplitude of g:1

a) Find the values of a and b

$a = 1 \qquad b = 1$

b) For which values of x *is* $f(x) = g(x)$ *if* $x \in [0^\circ; 360^\circ]$

$x = 45^\circ \qquad x = 225$

TRIG GRAPHS WORKSHEET (2)

1. The following graphs have been drawn below:

$f(x) = \tan x$ *and* $g(x) = a\cos x$

a) Determine the value of a.

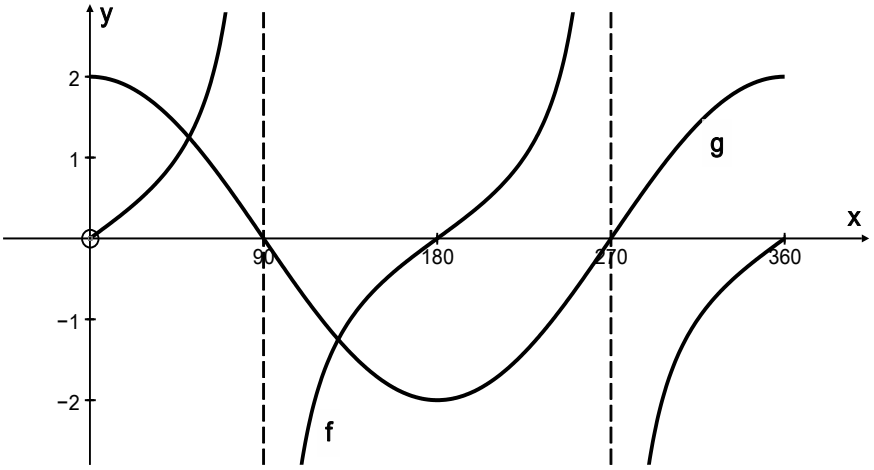
$a = 2$

b) If f(x) is shifted 2 units upwards, it represents the function h. State the <u>defining equation</u> of h.

$h(x) = \tan x + 2$

c) Write down the values of x for which g(x) – f(x) = 2

$x = 0$ $x = 360$



d) Write down the <u>range</u> of g.

$y \in [-2;2]$

e) Write down the <u>period</u> of f.

360°

2. The diagram below represents the graphs of f(x) and g(x) = tan θ

a) Give the <u>equation</u> of f(x)

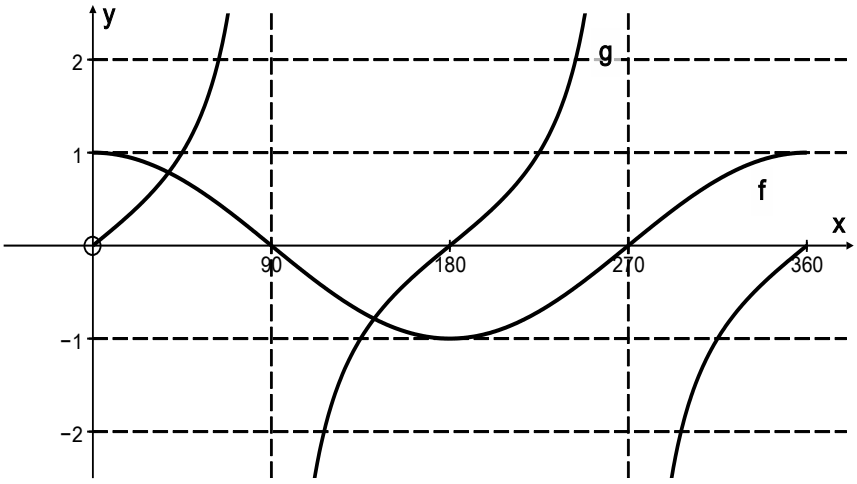
$f(x) = \cos x$

b) What is the period of g(x)?

180°

c) Write down the equations of the two lines that are the <u>asymptotes</u> of g(x) = tan

$x = 90$ $x = 270$



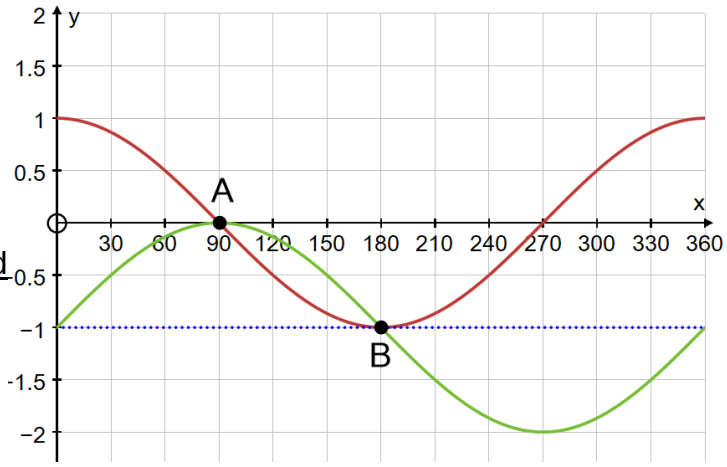
d) What is the <u>range</u> of g.

$y \in \mathbb{R}$

e) For what values of x is f(x) – g(x) = 1?

$x = 0$ $x = 360$

3. The diagram alongside shows the graph of f



a) What is the equation of f ?

f(x) = $\cos x$

b) Write down the amplitude and period of f

Amplitude: 1

Period: 360°

c) On the same set of axes draw the graph of g(x) = sinx – 1. Show clearly all the intercepts and turning points of both curves.

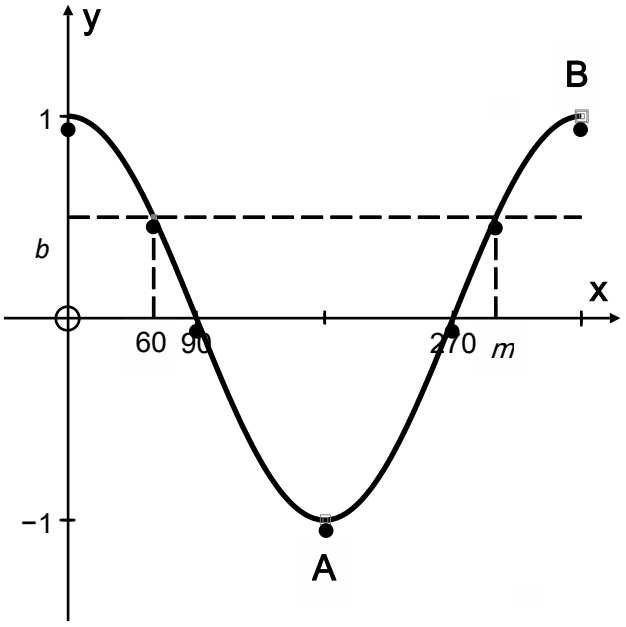
d) Label the points of intersection of both graphs f and g , using capital letters A, B, ...

e) Write down the solution of the equation cos x = sinx – 1 for $x \in [0^\circ ; 360^\circ]$

$x = 90$ $x = 180$

TRIG GRAPHS WORKSHEET (3)

1. The sketch $f(x)=\cos x$ is given



1.1. Determine the co-ordinates of A and B

$A(180;-1)$ $B(360;1)$

1.2. What is the value of m?

$m = 270 + 30 = 300$

1.3. Find the value of *b* (use a calculator)

$b = \cos 60$

$b = \frac{1}{2}$

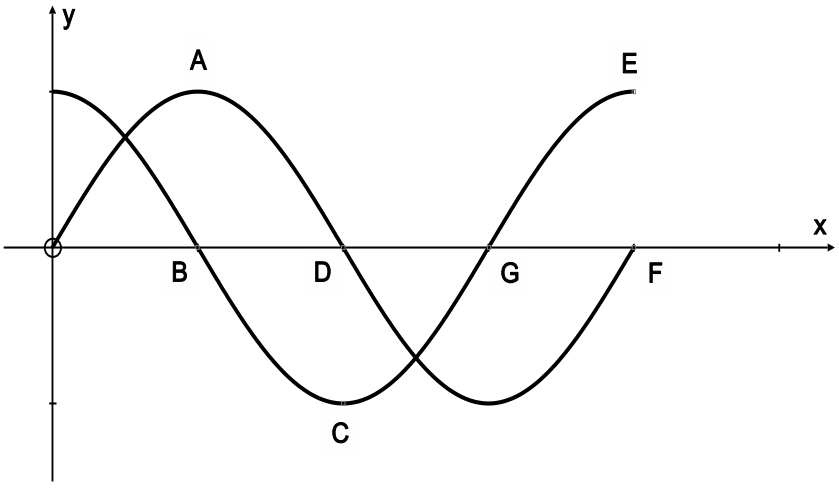
1.4. Use the graph to solve $\cos x + 1 = 0$

$\cos x = -1$

$x = 180$

2. The figure represents:

$h(x)=\sin\theta$ *and* $g(x) = \cos\theta$



2.1. Give the co-ordinates of:

A $(90;1)$
B $(90;0)$
C $(180;-1)$
D $(180;0)$
E $(360;1)$
F $(360;0)$

2.2. What is the minimum value of h?

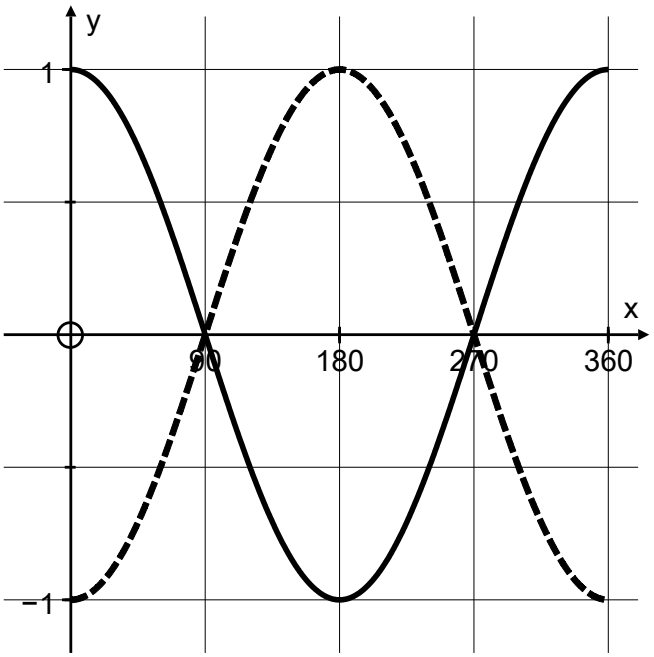
-1

2.3. For which values of θ is $\sin\theta.\cos\theta < 0$?

$x \in (90°;180°) \cup (270°;360°)$

3. The graph of $y = \cos x,\ 0^\circ \le x \le 360^\circ$ is shown.

Some transformations were performed to produce the dotted curve.



3.1. What is the period of $y = \cos x$?

360

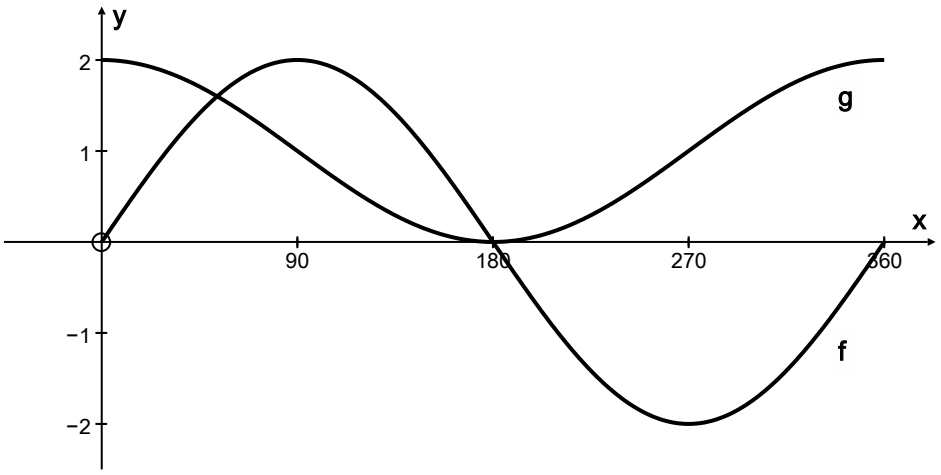
3.2. Write down the equation that describes the dotted line.

$y = -\cos x$

3.3. If the dotted line is moved 2 units downwards, write down the equation for the new curve.

$y = -\cos x - 2$

4. The diagram below represents the graphs of the functions $f(x) = a\sin x$ *and* $g(x) = \cos x + q$



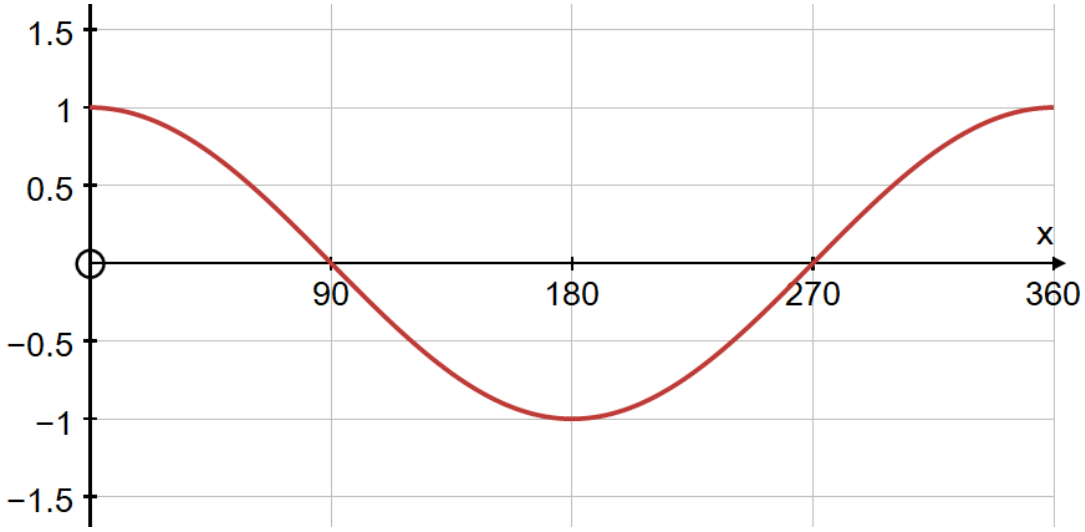
4.1. What is the value of a? $a = 2$

4.2. Write down the value of q $q = 1$

4.3. What is the range of g? $y \in [0;2]$

4.4. What is the value of f(270°)? -2

5. Draw a graph of $y = \cos x$ *for* $x \in [0^\circ ; 360^\circ]$



5.1. What is the amplitude of the graph?

1

5.2. What is the period of the graph?

360°

5.3. For what values of x is $\cos x = \frac{1}{2}$?

$x = 60^\circ$ $x = 300^\circ$

5.4. For what values of x is $\cos x \leq 0$?

$x \in [90;270]$

5.5. What does the equation of $y = \cos x$ become if the graph is shifted 3 units down?

$y = \cos x - 3$